

Woo Junhui

Data-Driven Control Research

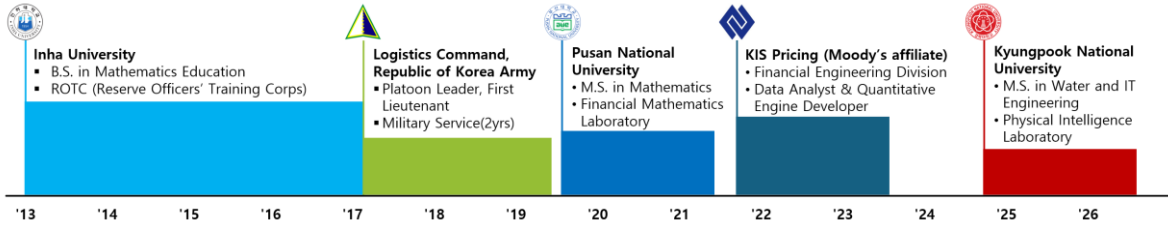
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Education & Experience



Quantitative Developer / Data Analyst KIS Pricing, Financial Engineering Group

Dec. 2021 – Aug. 2023

- Developed and maintained derivatives pricing modules and managed market data and numerical computation databases.
- Built automated ETL pipeline for KOSPI / HSCEI options market data using Python (Celery, APScheduler), reducing annual data cost by ~£8,268.
- Replaced Excel VBA-based Log-Normal Mixture volatility model with SABR-based Python automation systems, reducing daily validation workflow from 5 hours to 30 minutes (~90% reduction)

Achievement

- 2 Master's Degree** : Master of Engineering(2026) / Master of Science(2021)
- 3 Certifications** : Engineer Information Processing(2024) / ADsP(2024) / SQLD(2024)
- 3 Awards** : Academic Award(2020) / Military Commendation(2019) / Military Commendation(2017)

Publications (4 Journal Paper · 2 Conference Proceedings)

- [1] J. Woo, S. Lee*, "Error-Driven Reasoning Intelligence for Imitation Learning Beyond Demonstrator Performance with Certified Contractivity," *IEEE Transactions on Cybernetics*, Under Review, May 2026.
- [2] J. Woo, T. Kim, S. Lee*, "Enhancing Trajectory Tracking via Neural ODE-Based MPC with One-Shot Learned Error Dynamics," *Int. J. Robust Nonlinear Control*, Under Review, March 2026.
- [3] J. Woo, T. Kim, S. Lee*, "Neural ODE Predictive Control with Error Dynamics Learned from Demonstration for Trajectory Tracking," *Proc. IFAC World Congress 2026*, Accepted.
- [4] J. Woo, S. Lee*, "Nonlinear MPC with RNN-Based Neural ODEs Trained on Trajectory Tracking Demonstrations," *Proc. ICCAS 2025*, pp. 368-373.
- [5] D. Kim, J. Woo, J.-H. Yoon*, "Pricing American Lookback Options under a Stochastic Volatility Model," *Bull. Korean Math. Soc.*, vol.60, no.2, pp.361-388, 2023.
- [6] J. Woo, D. Kim, J.-H. Yoon*, "The Pricing of Vulnerable Options under a Constant Elasticity of Variance Model," *J. Chungcheong Math. Soc.*, vol.33, no. 2, pp.181-195, 2020.

Conference Presentations (2 International · 2 Domestic)

- [1] "Data-Driven LPV-MPC for Manipulator Tracking via Neural ODE-based Error Dynamics Learned from Demonstrations," *Ubiquitous Robots 2026 (UR2 026)*, Ritsumeikan University, Ibaraki, Osaka, Japan, Poster Presentation, July 2026. (Accepted, date TBD)
- [2] "Nonlinear MPC Using LSTM-Based Neural ODEs Trained on Trajectory Tracking Demonstrations," *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS 2025)*, Hangzhou, China, Poster Presentation, October 22, 2025.
- [3] "NeuralODE-LSTM 기반 오차 동역학 모델을 활용한 비선형 MPC에 의한 모바일 로봇 경로 추적" *The 56th Summer Conference of the Korea Institute of Electrical Engineering (KIEE 2025)*, Busan, Republic of Korea, Oral Presentation, July 18, 2025.
- [4] "금융시장지수의 국내 주력매매시장 적용가능성 연구," *Annual Meeting of the Korean Society for Industrial and Applied Mathematics (KSIAM 2019)*, Yeosu, Republic of Korea, Poster Presentation, November 9, 2019.

Academic Projects

Project 1 : Imitation Learning Beyond Demonstrator Performance with Certified Contractivity

Feb. 2026 – Jun. 2026

- ✓ Learned imitation policy from a single demonstration with ESO+PID demonstrator on a 4-DOF manipulator, with formal proof and numerical verification of closed-loop contractivity and ISS stability
- ✓ Validated the proposed framework on a 3-DOF manipulator using kinematic information against a PD demonstrator, where a single model trained on 50 repeated demonstrations per trajectory (Ellipse, Lemniscate, Rectangle) conforms generalization to highly nonlinear coupled systems.
- ✓ 91.8% (3-DOF) and 98.4% (4-DOF) tracking RMSE reduction over demonstrator

Project 2 : Enhancing Trajectory Tracking via Neural ODE-Based MPC with One-Shot Learned Error Dynamics

Apr. 2025 – Feb. 2026

- ✓ Single 10-min straight-line dataset with PE excitation generalizes across six distinct trajectories, all surpassing PD demonstrator under uncertain simulation environment.
- ✓ Sim-to-real transfer validated on ROBOTIS TurtleBot3 Burger via LMI-based terminal constraints derived from dissipativity-decreasing inequality
- ✓ Distance RMSE reduced by 70.4% / 72.3% / 75.1% and Heading RMSE by 60.1% / 64.1% / 52.0% across Lemniscate / Perturbed Circle / Perturbed Spiral trajectories.

Project 3 : ARX-ESO-Based MPC and Efficiency Optimization for Industrial Thermal Processes

Nov. 2025 – Dec. 2025

- ✓ Analyzed client-provided PID control logs and operation datasets, designed and implemented ARX(1)+ESO-based MPC tailored to a two-stage industrial heating process, and delivered a technical report with implementation guide.
- ✓ Generalized across 6 independent operating datasets with varying initial conditions(557°C - 685°C) from a single identified model.
- ✓ Average process time reduced by ~47% (490→336 min), total cost reduced by ~51%, heater effort reduced by ~16%, IAE reduced by ~31%, steady-state convergence within $\pm 5^\circ\text{C}$ satisfied across all scenarios.



- Numerical Simulation
- Model design & Training
- API Integration
- Workflow Automation

- Database management
- Schema design
- Data manipulation via SQL

- Data collection & validation in Gazebo & RViz
- Real robot Control & validation
- Design Robot with URDF

- Static web design
- Academic writing
- Structured prompting for LLM workflows